01 Hydrogen and Its Compounds

Introduction:

Hydrogen is the lightest element in the periodic table. It is the lightest non-metal. It is the simplest element in periodic table having only $1e^{\ominus}$, 1p & no neutron.

Electronic configuration of H is $1s^1$

Number of $e^{\ominus}$ = 1 Number of orbitals = 1

Number of shells = 1 Number of subshells =1

It is discovered by Henry Cavendish & it was called inflammable element.

The name hydrogen was given by Lavoisier

Position of Hydrogen in The Periodic Table

Hydrogen is the first element in the periodic table. However, its placement in the periodic table has been a subject of discussion in the past. As you know by now that the elements in the periodic table are arranged according to their electronic configurations. Hydrogen has electronic configuration $1s^1$. On one hand, its electronic configuration is similar to the outer electronic configuration (ns^1) of alkali metals, which belong to the first group of the periodic table. On the other hand, like halogens (with ns^2np^5 configuration belonging to the seventeenth group of the periodic table), it is short by one electron to the corresponding noble gas configuration, helium ($1s^2$). Hydrogen, therefore, has resemblance to alkali metals, which lose one electron to form unipositive ions, as well as with halogens, which gain one electron to form uni-negative ion. Like alkali metals, hydrogen forms oxides, halides and sulphides. However, unlike alkali metals, it has a very high ionization enthalpy and does not possess metallic characteristics under normal conditions. In fact, in terms of ionization enthalpy, hydrogen resembles more with halogens, $\Delta_i H$ of Li is 520 kJ mol^{-1}, F is 1680 kJ mol^{-1} and that of H is 1312 kJ mol^{-1}. Like halogens, it forms a diatomic molecule, combines with elements to form hydrides and a large number of covalent compounds. However, in terms of reactivity, it is very low as compared to halogens.

Inspite of the fact that hydrogen, to a certain extent resembles both with alkali metals and halogens, it differs from them as well. Now the pertinent question arises as where should it be placed in the periodic table? Loss of the electron from hydrogen atom results in nucleus (H^+) of $\sim 1.510^{-3}$ pm size. This is extremely small as compared to normal atomic and ionic sizes of 50 to 200 pm. As a consequence, H^+ does not exist freely and is always associated with other atoms or molecules. Thus, it is unique in behaviour and is, therefore, best placed separately in the periodic table.

Isotopes of Hydrogen:

Hydrogen has three isotopes: **protium**, ^1_1H , **deuterium**, ^2_1H or D and **tritium**, ^3_1H or T. These isotopes differ from one another in respect of the presence of neutrons. Ordinary hydrogen, protium, has no neutrons, deuterium (also known as heavy hydrogen) has one and tritium has two neutrons in the nucleus. In the year 1934, an American scientist, Harold C. Urey, got Nobel Prize for separating hydrogen isotope of mass number 2 by physical methods.

The predominant form is protium. Terrestrial hydrogen contains 0.0156% of deuterium mostly in the form of HD. The tritium concentration is about one atom per 1018 atoms of protium. Of these isotopes, only tritium is radioactive and emits low energy β^- particles.

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Atomic and Physical Properties of Hydrogen

Property	Protium(H) or ordinary hydrogen	Deuterium(D) or heavy hydrogen	Tritium(T)
Atomic number	1	1	1
Mass number	1	2	3
Exact atomic mass	1.008123	2.0142	3.0170
Symbol	$^{1}_{1}H$	$^{2}_{1}H$ or $^{2}_{1}D$	$^{3}_{1}H$ or $^{3}_{1}T$
Molecular formula	H_2	D_2	T_2
No. of protons in the nucleus	1	1	1
No. of neutrons in the nucleus	Nil	1	2
No. of electron	1	1	1
Electronic configuration	$1s^1$	$1s^1$	$1s^1$
Relative abundance	99.984%	0.016%	10^{-15}%
Stability	Stable	Stable	Unstable (Radioactive)

Property	H_2	D_2	T_2
Heat of fusion & vaporisation	Minimum	H < D < T	Maximum
Relative abundance (%)	99.985	0.0156	10^{-15}
Relative atomic mass (g mol^{-1})	1.008	2.014	3.016
Melting point / K	13.96 (–259 °C)	18.73 (–254.3 °C)	20.62 (–252.4 °C)
Boiling point /K	20.39 (–252.6 °C)	23.67 (–249.3 °C)	25.0 (–248 °C)
Density / gL^{-1}	0.09	0.18	0.27
Enthalpy of fusion/KJ mol^{-1}	0.117	0.197	–
Enthalpy of vaporization/kJ mol^{-1}	0.904	1.226	–
Ionization enthalpy/kJ mol^{-1}	1312	–	–
Electron gain enthalpy/kJ mol^{-1}	–73	–	–
Covalent radius/pm	37	–	–
Ionic radius(H^-)/pm	208		

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Since the isotopes have the same electronic configuration, they have almost the same chemical properties. The only difference is in their rates of reactions, mainly due to their different enthalpy of bond dissociation. However, in physical properties these isotopes differ considerably due to their large mass differences.

Illustration 1

Which of the following reaction is fast & why ?

(i) $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$

(ii) $CD_4 + Cl_2 \longrightarrow CD_3Cl + DCl$

Solution:

(i) because C–H bond energy is less in comparison to C–D bond energy.

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Different forms of Hydrogen:

(a) Based on oxidation Number

There are three types of hydrogen

	$H^{\oplus}$	$H^{\ominus}$	H
	Proton	Hydride	Atomic hydrogen
Number of electron	0	2	1
Oxidation number	+1	–1	0
Formation	$H \rightarrow H^{\oplus} + e^{\ominus}$	$H + e^{\ominus} \rightarrow H^{\ominus}$	$H_2 \xrightarrow{\Delta} 2H$

Note :

In the aqueous state proton ($H^{\oplus}$) exist as $H^{\oplus}(H_2O)_n$

Where n is a large number.

If $n = 1 \longrightarrow H_3O^{\oplus}\ [H^{\oplus}(H_2O)_1]$

$n = 2 \longrightarrow H^{\oplus}(H_2O)_2$ or $O_2H_5^{\oplus}$

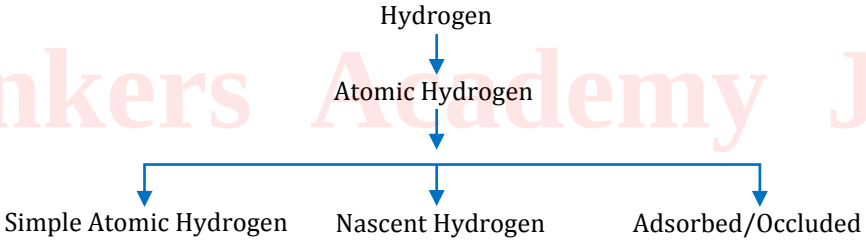
Illustration 2

In the aqueous solution hydrogen ion exist as

(A) $H_3O^{\oplus}$ (B) $H^{\oplus}(H_2O)_2$ (C) $H^{\oplus}(H_2O)_n$ (D) All

Ans. (D)

(b) Based on reactivity :



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Atomic hydrogen :

(i) Simple atomic hydrogen – To obtained Atomic hydrogen, molecular hydrogen is passed through an electric arc struck between Tungsten metal electrodes at 3000-5000 K.

$$H_2 \underset{}{\overset{3000\text{-}5000\,K}{\rightleftharpoons}} 2H$$

→ Favourable condition – Favourable condition are high temp & low pressure.

→ The life period of atomic hydrogen is $\left(\frac{1}{3}\right)^{rd}$ of a second but can be extended to 10 seconds with some condition.

(ii) Nascent hydrogen – Hydrogen at the moment of its birth it called nascent hydrogen means which forms at the instance is known as Nascent hydrogen.

It is formed only by some specific chemical reaction.

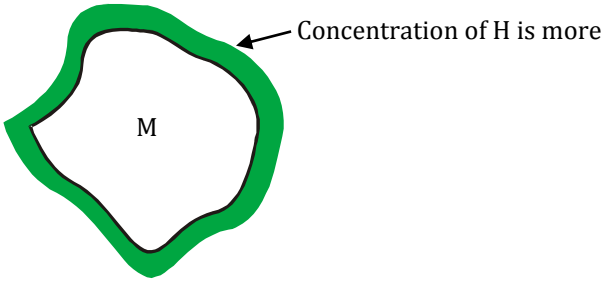
(a) Acid + Metals $Zn + H_2SO_4 \longrightarrow ZnSO_4 + 2H$

(b) Base + amphoteric metal (Be, Zn, Al, Ga, Pb, Sn) $2NaOH + Be \longrightarrow Na_2BeO_2 + 2H$

(c) C_2H_5OH + Alkali metal $C_2H_5OH + Na \longrightarrow C_2H_5ONa + H$

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(iii) Adsorbed/Occluded hydrogens



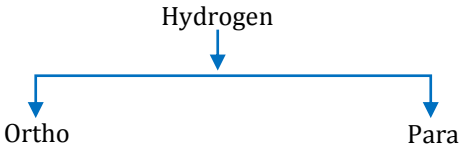
Adsorbed H is hydrogen present at the outer surface of metal.

Occlusion – The property of metal to adsorb any gas on its surface is called occlusion.

Some metals such as Pd, Pt, Au, Fe, Ni etc. adsorbs hydrogen on their surface.

Reactivity order - Atomic hydrogen > Nascent hydrogen > Adsorbed hydrogen

(iii) Based on Nuclear spin (Nuclear isomers)



(a) Ortho hydrogen – The molecular form of hydrogen having same spin of proton is called ortho hydrogen.

(b) Para hydrogen – The molecular form of hydrogen having opposite spin of proton is called para hydrogen.

In ortho hydrogen spin of proton is same, so they will repel each other & because of this repulsion, internal energy of ortho hydrogen increases. So ortho hydrogen has more internal energy.

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Stability of ortho & para hydrogen :

Stability of ortho & para hydrogen depends upon temperature condition.

At low temp : para hydrogen is more stable than ortho hydrogen while at high temp ortho hydrogen is more stable than para hydrogen.

$$P_{H_2} \underset{T\downarrow}{\overset{T\uparrow}{\rightleftharpoons}} O_{H_2}$$

	Para	Ortho
At 25°C	75%	25%
At (–253°C)/20K	0	100%

Note :

(i) We can obtain 100% pure para hydrogen at low temp but can't obtain 100% pure ortho hydrogen because at high temp parahydrogen will dissociate into atomic hydrogen.

(ii) Ortho & Para hydrogen differs only in physical properties but have same chemical properties.

Method of preparation of Dihydrogen(H_2) (Lab preparation methods):

1. **By alkalies :** Only (Be, Zn, Al, Sn, Pb, Si) (Amphoteric metal) react with boiling NaOH or KOH they evolve H_2.

$$Be + 2NaOH \longrightarrow Na_2BeO_2 + H_2\uparrow$$
(sodium beryliate)

$$2Al + 2NaOH + 2H_2O \longrightarrow 2NaAlO_2 + 3H_2\uparrow$$
(sodium meta aluminate)

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$Zn + 2NaOH \longrightarrow Na_2ZnO_2 + H_2\uparrow$
(sodium zincate)
$Si + 2NaOH + H_2O \longrightarrow Na_2SiO_3 + 2H_2\uparrow$
(sodium silicate)
$Sn + 2NaOH + H_2O \longrightarrow Na_2SnO_3 + 2H_2\uparrow$
(sodium stannate)
$Pb + 2NaOH + H_2O \longrightarrow Na_2PbO_3 + 2H_2\uparrow$
(sodium plumbate)

2. **From acids :** The metal which are placed above H_2 in electrochemical series when react with dilluted acids they liberate H_2.

e.g. $Fe + H_2SO_4 \longrightarrow FeSO_4 + H_2$
(dil.)
$Cu + H_2SO_4 \longrightarrow \times$ (No reaction)

Lab preparation : When impure granulated Zn reacts with dil. H_2SO_4 it forms H_2

$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$
(impure) (dil)

H_2 bubbles give popping sound when burning candle is introduced to them.

Illustration 3

Why we use impure Zn, in place of pure Zn.

Solution:

Because the rate of reaction with pure Zn is very slow.

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3. **From water :**

All the metals which are placed above than H_2 in electrochemical series when react with water they evolve H_2.

$Zn + H_2O \longrightarrow ZnO + H_2\uparrow$

Three types of water is used

(i) Cold water : The temperature of cold water is 7 to 25°C this water is used for highly reactive metals.

Such as Li, K, Ba, Sr, Ca, Na, means alkali metals and alkaline earth metals.

The reactions with alkali metals are vigorous to minimum the rate of reaction these metals are used in the form of amalgam [Na.Hg].

$2Na.Hg + 2H_2O \longrightarrow 2NaOH + H_2\uparrow$

(ii) Hot water : The temperature of hot water is 25°C to 90°C. This water is used for reactive metals, such as Mg, Al, Mn, Zn, Cr.

(iii) Steam : The temperature of steam is more than 100°C. This form of water is used for very less reactive metals like Fe, Cd, Co, Ni, Sn, Pb.

Condition for best yield of H_2.

(i) Cold water $\longrightarrow$ With highly reactive metals.
(ii) Hot water $\longrightarrow$ With reactive metals
(iii) Steam $\longrightarrow$ With less reactive metals.

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Method to prepare pure Dihydrogen gas (H_2):

Pure hydrogen (purity > 99.95 %) required for military purposes and can be produced on the spot as follows :

1. **For highly pure hydrogen (purity > 99.95 %), H_2 is obtained by electrolysis of warm aqueous barium hydroxide solution using Nickel electrodes.**

2. **By reaction of pure dil. H_2SO_4 with Mg Ribbon :**
   $Mg + H_2SO_4 \longrightarrow MgSO_4 + H_2$
   (dil.)

3. **By reaction of NaH (Ionic hydride) with water :**
   Whenever ionic hydride reacts with water it form H_2.
   $LiH + H_2O \longrightarrow LiOH + H_2$
   $NaH + H_2O \longrightarrow NaOH + H_2$
   $CaH_2 + 2H_2O \longrightarrow Ca(OH)_2 + H_2$
   (Hydrolith) (lime water)
   $NaBH_4 + H_2O \longrightarrow NaBO_2 + 4H_2$
   (Sodium borohydride) (Sodium metaborate)

4. **Uyeno method :** This method is used for military purpose because by this reaction we can prepare H_2 in a rapid manner by heating Al scrap with KOH solution.
   $2Al + 2KOH + 2H_2O \longrightarrow 2KAlO_2 + 3H_2$
   (Potassium meta aluminate) (Rapidly prepared hydrogen gas)

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Commercial or industrial method to prepare H_2:

1. **Electrolysis of water (commercial method) :** Electrolysis of impure or acidified water (i.e. having 15-20% solution of alkali or acid) is done to prepare pure hydrogen using inert solids like platinum/palladium as electrodes.
   $4H_2O \rightleftharpoons 4H^{\oplus} + 4OH^{\ominus}$
   at cathode $4H^{\oplus} + 4e^{\ominus} \longrightarrow 2H_2$
   at anode $4OH^{\ominus} \longrightarrow 2H_2O + O_2 + 4e^{\ominus}$
   The $SO_4^{2\ominus}$ or $K^{\oplus}$ ion present in acid or alkali does not move towards anode or cathode as their discharge potential is higher than of $OH^{\ominus}$ ions or $H^{\oplus}$ ions respectively.

2. **Lane process :** Steam is passed over hot iron, it get converted into Fe_3O_4 & H_2.
   $3Fe + 4H_2O \xrightarrow{1023-1073\,K} Fe_3O_4 + 4H_2\uparrow$
   Iron is regenerated by reducing Fe_3O_4 into Fe by water gas (CO + H_2)
   $2Fe_3O_4 + 4CO + 4H_2 \longrightarrow 6Fe + 4CO_2 + 4H_2O$
   $Fe_3O_4 + 2CO + 2H_2 \longrightarrow 3Fe + 2CO_2 + 2H_2O$
   This process is a continuous process.

3. **Water – gas shift reaction or Bosch Process :**
   Reaction of steam on hydrocarbons or red hot coke at high temperatures in the presence of catalyst yields hydrogen.
   $C_nH_{2n+2} + nH_2O \xrightarrow[Ni]{1270\,K} nCO + (2n + 1)\,H_2$

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e.g.,

$$CH_4(g) + H_2O(g) \xrightarrow[Ni]{1270\,K} \underbrace{CO(g) + 3H_2(g)}_{\text{(water-gas)}}$$

The mixture of CO and H_2 is called water gas. As this mixture of CO and H_2 is used for the synthesis of methanol and a number of hydrocarbons, it is also called *synthesis gas or* **'syngas'**. Nowadays 'syngas' is produced from sewage, saw-dust, scrap wood, newspapers etc. The process of producing 'syngas' from coal is called *'coal gasification'*.

$$C(s) + H_2O(g) \xrightarrow[Ni]{1270\,K} \underbrace{CO(g) + H_2(g)}_{\text{(water-gas/syngas)}}$$

The production of dihydrogen can be increased by reacting carbon monoxide of syngas mixtures with more steam in the presence of iron chromate (Fe_2CrO_4 / Fe_2O_3 and Cr_2O_3) as a catalyst.

$$CO(g) + H_2(g) + H_2O(g) \xrightarrow[\text{catalyst}]{673K} CO_2(g) + H_2(g)$$ [Water gas shift reaction]

This is called water-gas shift reaction /**Bosch process**. Carbon dioxide is removed by scrubbing with sodium arsenite solution. Presently ~77% of the industrial dihydrogen is produced from petro-chemicals, 18% from coal, 4% from electrolysis of aqueous solutions and 1% from other sources.

Physical Properties of Dihydrogen:

(i) Hydrogen is a lightest, colourless, odourless and tasteless gas. It is sparingly (very less) soluble/almost insoluble in water due to its non-polar nature. It is inflammable and less reactive gas.

(ii) Its freezing point (– 259.2°C) and boiling point (–252°) are very low indicating less intermolecular attraction. Due to low freezing point liquid hydrogen is used as a **cryogenic fluid** (to produce low temperature).

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(iii) H—H bond energy [104 Kcal mol^{-1}] and 436 KJ/mole

(iv) H – H bond length [74 pm] so H_2 is less reactive and require high temp for reaction.

(v) It can be liquefied under low temperature and high pressure.

Chemical properties of H_2:

(i) H_2 is neutral in nature that why it does not react with acids & bases.

(ii) **Less reactive:** Hydrogen is very less reactive in nature because of very high bond dissociation enthalpy.

(iii) **Combustible nature :** H_2 is highly combustible in nature & it burns with oxygen or air with pale blue flame to give water. But it doesn't support combustion like O_2 does.

$$2H_2 + O_2 \longrightarrow 2H_2O(l);\ \Delta H^{\ominus} = -285.9\ kJ/mole$$

(iv) Reaction with highly electropositive metals :

IA & IIA group elements are called highly electropositive metal. Whenever H reacts with these metals they form ionic hydrides.

$$2Na + H_2 \longrightarrow 2NaH$$

$$Ca + H_2 \longrightarrow \underset{\text{(hydrolith)}}{CaH_2}$$

(v) Reaction with non-metal :

(a) Reaction with halogen-forms hydrogen halides

$$H_2 + X_2 \longrightarrow 2HX$$

[$X_2 = F_2, Cl_2, Br_2, I_2$]

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order of reactivity of halogen with hydrogen

$F_2 > Cl_2 > Br_2 > I_2$

While the reaction with fluorine occurs even in the dark, with iodine it requires a catalyst.

(b) Reaction with carbon :

$2H_2 + C \xrightarrow{1420K} CH_4$

(Methane)

(c) Reaction with sulphur

$H_2 + S \longrightarrow H_2S$

(d) With nitrogen – (Preparation of ammonia by Haber's process)

$N_2 + 3H_2 \xrightarrow[\text{Fe, Mo, 200 atm}]{683\ K} 2NH_3;\ \Delta H^{\ominus} = -92.6\ kJ/mole$

(Fe as catalyst, Mo as promoter)

(vi) Reducing nature :

$CuO + H_2 \longrightarrow Cu + H_2O$

(vii) Reaction with carbon mono oxide (Preparation of Methanol) :

$2H_2 + CO \xrightarrow[373k]{ZnO/Cr_2O_3} CH_3OH$ (Methanol)

(viii) Hydrogenation of unsaturated hydrocarbon :

$C_2H_2 + 2H_2 \longrightarrow C_2H_6$

$C_2H_4 + H_2 \xrightarrow[473k]{Ni/Pd/Pt} C_2H_6$

When unsaturated hydrocarbon having C = C (Alkene) or C ≡ C (Alkyne) reacts with hydrogens in the presence of finely divided Ni/Pd/Pt forms saturated hydrocarbons.

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(a) Hydrogenation of vegetable oil.

Vegetable oils are also called polyunsaturated oils because they contain many C=C bond. When these oil are exposed to air for a long time then double bond will get oxidized and the oil becomes. Rancid (having foul. smell or unpleasant taste) in nature. So, to avoid this, vegetable oils are converted into edible fats (Vanaspati Ghee).

Vegetable oils + $H_2 \xrightarrow{Ni}$ edible fats solid

(liquid)

This whole process is known as hydrogenation or hardening of oil.

(b) Hydroformylation of olefins

Hydroformylation of olefins yields aldehydes which further undergo reduction to give alcohols.

$H_2 + CO + RCH = CH_2 \rightarrow RCH_2CH_2CHO$

$H_2 + RCH_2CH_2CHO \rightarrow RCH_2CH_2CH_2OH$

Uses of Dihydrogen(H_2):

1. The largest single use of dihydrogen is in the synthesis of ammonia which is used in the manufacture of nitric acid and nitrogenous fertilizers.
2. Dihydrogen is used in the manufacture of Vanaspati fat by the hydrogenation of polyunsaturated vegetable oils like soya bean, cotton seeds etc.
3. It is used in the manufacture of bulk organic chemicals, particularly methanol.

   $CO(g) + 2H_2(g) \xrightarrow[\text{catalyst}]{\text{cobalt}} CH_3OH(l)$
4. It is widely used for the manufacture of metal hydrides.
5. It is used for the preparation of hydrogen chloride, a highly useful chemical.

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6. In metallurgical processes, it is used to reduce heavy metal oxides to metals.
7. Atomic hydrogen and oxy-hydrogen torches find use for cutting and welding purposes. Atomic hydrogen atoms (produced by dissociation of dihydrogen with the help of an electric arc) are allowed to recombine on the surface to be welded to generate the temperature of 4000 K.
8. It is used as a rocket fuel in space research.
9. Dihydrogen is used in fuel cells for generating electrical energy. It has many advantages over the conventional fossil fuels and electric power. It does not produce any pollution and releases greater energy per unit mass of fuel in comparison to gasoline and other fuels.

Hydrides:

Dihydrogen, under specific conditions combines with almost all elements except noble gases to form binary compounds called hydrides.

These are of three types :

(1) Ionic/Salt like/Saline hydrides :

Compounds of hydrogen with s-block elements except berylium & magnesium are called ionic hydrides.

LiH, NaH, KH, RbH, CsH, CaH_2, SrH_2, BaH_2

BeH_2, MgH_2 are covalent polymeric hydride.

- Structure of these hydrides are similar to rock salt, so they are also called salt like/saline hydrides.
- Down the group size↑ Lattice energy↓ stability↓ Melting point↓ Boiling point↓
- On electrolysis of these hydrides, hydrogen is liberated at anode.
- These hydrides form complex hydrides which are very good reducing agents.

$$4LiH + AlCl_3 \longrightarrow LiAlH_4 + 3LiCl$$

$$4NaH + \underbrace{B(OCH_3)_3}_{Me_3BO_3\text{(Trimethyl borate)}} \rightarrow NaBH_4 + \underset{\text{(Sodium methoxide)}}{3NaOCH_3}$$

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$NaBH_4 \longrightarrow$ Sodium borohydride

$LiAlH_4 \longrightarrow$ Lithium aluminium hydride.

- Saline hydrides react violently with water producing dihydrogen gas.

$$NaH_{(s)} + H_2O_{(l)} \longrightarrow NaOH_{(aq.)} + H_{2(g)}$$

- Lithium hydride is rather unreactive at moderate temperatures with O_2 or Cl_2. It is, therefore, used in the synthesis of other useful hydrides, e.g.,

$$8LiH + Al_2Cl_6 \rightarrow 2LiAlH_4 + 6LiCl$$

$$2LiH + B_2H_6 \rightarrow 2LiBH_4$$

(2) Metallic / Interstitial hydrides :

They are the compounds of d & f-block elements.

In these hydrides hydrogen occupies interstitial sites present in metallic lattice, so they are called interstitial hydrides.

- Properties of these hydrides are similar to parent metals, so they are also known as metallic hydrides.
- These hydrides are non. stochiometric in nature (i.e. having variable composition in non whole number ratio).

  ZrH_X (x = 1.3 – 1.75)

  TiH_X (x = 1.8 – 2)
- Metals of group 7,8,9 do not form any hydrides so this particular part of periodic table is known as **hydride gap**.
- Chromium exceptionally forms hydride (CrH) among all group VI elements.

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(3) Covalent/Molecular hydrides

- These are the compounds of hydrogen with p-block elements (except noble gases) and some s-block elements (Be and Mg) CH_4, NH_3, H_2O, HF, etc.
- These hydrides exist as discrete molecules, due to weak Van derwaals force of attraction among molecules, so they are also known as molecular hydrides. These hydrides are non-conductor of electricity.
- Being covalent, they are volatile compounds.

These hydrides are again divides into 3 categories.

(a) Electron deficient hydrides :

- They are the hydrides of group 13 elements.
- BH_3, AlH_3, GaH_3 – In these hydrides central element does not have complete octet. i.e. that's why they are called electron deficient compounds.
- These hydrides exist in polymeric form example B_2H_6, B_4H_{10}, $(AlH_3)_n$ etc.
- Due to electron deficiency their behaviour is as Lewis acid.

(b) Electron precise hydrides –

- They are the hydrides of group 14 element. e.g. CH_4, SiH_4, GeH_4
- In these type of hydrides central elements has $8e^{\ominus}$ in their outer most shell.
- All such hydrides are tetrahedral.

(c) Electron rich hydrides :

- These are the hydrides of group 15, 16, 17 e.g. $\ddot{N}H_3, H_2\ddot{O}, H\ddot{F}:$ etc.
- In these hydrides lone pair are present on central atom which can be given to others.
- They act as Lewis base, So they are called electron rich hydrides.
- Hydrides of 2nd period elements of 15,16,17 group have high B.P due to intermolecular H-bonding than rest of the element hydrides of their respective group.

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Water (H_2O):

A major part of all living organisms is made up of water. Human body has about 65% and some plants have as much as 95% water. It is a crucial compound for the survival of all life forms. It is a solvent of great importance. The distribution of water over the earth's surface is not uniform. The estimated world water supply is given in Table.

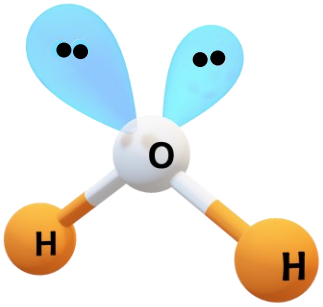
Estimated World Water Supply

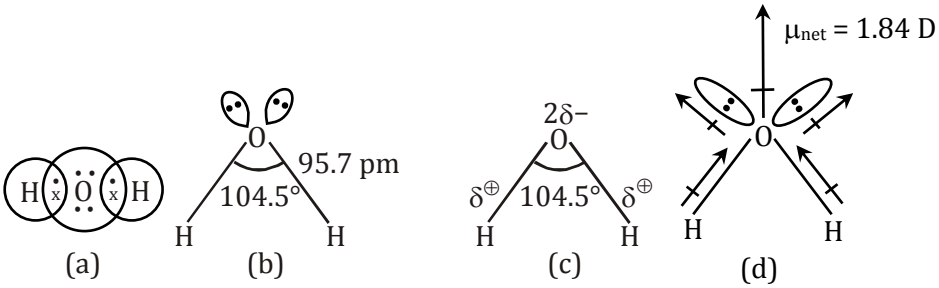
Source	% of total
Oceans	97.33
Saline lakes and inland seas	0.008
Polar ice and glaciers	2.04
Ground water	0.61
Lakes	0.009
Soil moisture	0.005
Atomspheric water vapour	0.001
Rivers	0.001

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- **Structure of Water**

In the gas phase water is a bent molecule with a bond angle of 104.5° and O–H bond length of 95.7 pm as shown in Fig (a).





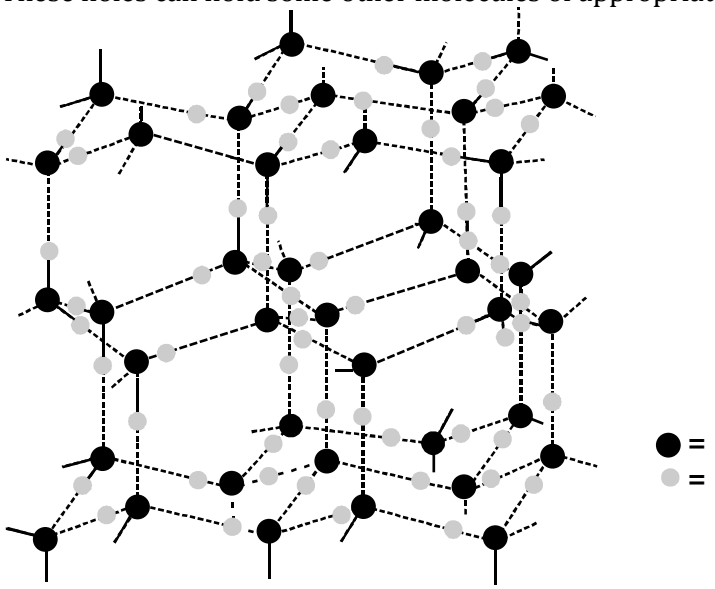
(a) Lewis dot structure of water molecule. (b) The bent structure of water; (c) the water molecule as a dipole and (d) the resultant dipole moment of water molecule.

It is a highly polar molecule, (Fig (b)). Its orbital overlap picture is shown in Fig. (c). In the liquid phase water molecules are associated together by hydrogen bonds.

The crystalline form of water is ice. At atmospheric pressure ice crystallises in the hexagonal form, but at very low temperatures it condenses to cubic form. Density of ice is less than that of water. Therefore, an ice cube floats on water. In winter season ice formed on the surface of a lake provides thermal insulation which ensures the survival of the aquatic life. This fact is of great ecological significance.

- **Structure of Ice**

Ice has a highly ordered three dimensional hydrogen bonded structure as shown in Fig. Examination of ice crystals with X-rays shows that each oxygen atom is surrounded tetrahedrally by four other oxygen atoms at a distance of 276 pm. Hydrogen bonding gives ice a rather open cage type structure with wide holes. These holes can hold some other molecules of appropriate size interstitially.



The Structure of Ice

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Properties of pure water :

(a) Physical properties :

(i) Pure water is colourless, tasteless and odourless. It gives bluish tinge in thick layers.

(ii) It freezes at 0°C and boils at 100°C.

(iii) Its maximum density is 1.00 at 4°C.

(iv) It is a polar molecule and has V-shaped structure. The bond angle is 104.5°.

(v) It has a high dielectric constant. The polar character of water makes it an excellent solvent for polar and ionic substances.

(vi) It is a poor conductor of electricity.

(vii) It has the tendency to associate in the liquid state and solid state it exists not as a single H_2O molecule but as associated molecules through hydrogen bonding. The existence of hydrogen bonding is responsible for high values of specific heat, the latent heat of fusion and latent heat of vaporisation, high freezing point, high boiling point, high melting point as compared to H_2S and H_2Se.

(viii) Boiling point of water is high due to the presence of hydrogen bonding

Physical Properties of H_2O and D_2O

Property	H_2O	D_2O
Molecular mass (g mol^{-1})	18.0151	20.0276
Melting point/K	273.0	276.8
Boiling point/K	373.0	374.4
Enthalpy of formation/kJ mol^{-1}	–285.9	–294.6
Enthalpy of vaporisation (373K)/kJmol^{-1}	40.66	41.61
Enthalpy of fusion/kJ mol^{-1}	6.01	-
Temp of max. density/K	276.98	284.2
Density(298K)/g cm^{-3}	1.0000	1.1059
Viscosity/centipoise	0.8903	1.107
Deielectric constant/$C^2/N.m^2$	78.39	78.06
Electrical conductivity (293K/ohm^{-1} cm^{-1})	5.7×10^{-8}	-

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The high heat of vaporisation and heat capacity are responsible for moderation of the climate and body temperature of living beings. It is an excellent solvent for transportation of ions and molecules required for plant and animal metabolism. Due to hydrogen bonding with polar molecules, even covalent compounds like alcohol and carbohydrates dissolve in water.

(b) Chemical properties :

(i) Water is neutral, amphoteric in nature. pH of the pure water is 7. It is a weak electrolyte and feebly (very less) ionises into $H^{\oplus}$ and $OH^{\ominus}$ ions.

$$H_2O \rightleftharpoons H^{\oplus} + OH^{\ominus}$$

In pure water $[H^{\oplus}] = [OH^{\ominus}] = 10^{-7}$ at 25°C.

(ii) **Reaction with metals:** It reacts with active metals and evolves hydrogen and act as an oxidising agent. The reaction is exothermic in the case of alkali and alkaline earth metals.

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2$$

$$Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2$$

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(iii) **Reaction with non metals :**

(a) Greenish- yellow chlorine decomposes cold water forming HCl and HClO.

$$Cl_2 + H_2O(cold) \rightarrow HCl + \underset{\text{(Hypochlorous acid)}}{HClO}$$

$$Cl_2 + H_2O(Hot) \rightarrow HCl + \underset{\text{(Chloric acid)}}{HClO_3}$$

(b) When steam is passed over red hot coke (1000°C), water gas is formed.

$$C(s) + H_2O(g) \xrightarrow[Ni]{1270\,K} \underbrace{CO(g) + H_2(g)}_{\text{(water–gas/syngas)}}$$

(iv) **Reaction with non-metallic oxides :** Acidic oxides combine with water to form acids.

$CO_2 + H_2O \rightleftharpoons H_2CO_3$ (Carbonic acid)

$SO_2 + H_2O \longrightarrow H_2SO_3$ (Sulphurous acid)

$SO_3 + H_2O \longrightarrow H_2SO_4$ (Sulphuric acid)

$P_2O_5 + 3H_2O \longrightarrow 2H_3PO_4$ (Orthophosphoric acid)

$N_2O_5 + H_2O \longrightarrow 2HNO_3$ (Nitric acid)

$Cl_2O_7 + H_2O \longrightarrow 2HClO_4$ (Perchloric acid)

(v) **Reaction with metallic oxides :** basic oxides combine with water to form alkalis or base.

$Na_2O + H_2O \longrightarrow 2NaOH$

$CaO + H_2O \longrightarrow Ca(OH)_2$

$2KO_2 + 2H_2O \longrightarrow 2KOH + H_2O_2 + O_{2(g)}\uparrow$

$Na_2O_2 + 2H_2O \longrightarrow 2NaOH + H_2O_2$

$Li_2O_2 + 2H_2O \longrightarrow 2LiOH + H_2O_2$

(vi) **Reaction with hydrides, carbides, nitrides, phosphides:** Water decomposes these compounds with liberation of hydrogen, acetylene (or methane), ammonia, phosphine respectively.

$CaH_2 + 2H_2O \longrightarrow Ca(OH)_2 + 2H_2$

$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2$

$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$

$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$

$AlN + 3H_2O \longrightarrow Al(OH)_3 + NH_3$

$Ca_3N_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2NH_3$

$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$

(vii) **Reaction with Chlorides:**

$PCl_5 + 4H_2O \longrightarrow POCl_3 + 2HCl \longrightarrow H_3PO_4 + 5HCl$

(viii) **Hydrates Formation:** From aqueous solutions many salts can be crystallised as hydrated salts. Such an association of water is of different types viz.,

(i) coordinated water e.g.,

$[Cr(H_2O)_6]^{3+}\,3Cl^-$

(ii) interstitial water e.g., $BaCl_2.2H_2O$

(iii) hydrogen-bonded water e.g.,

$[Cu(H_2O)_4]^{2+}\,SO_4^{2-}\cdot H_2O$ in $CuSO_4\cdot 5H_2O$,

Heavy water (D_2O):

Method of preparation :

Repeated electrolysis of H_2O : On multistage electrolysis of water (impure) H_2O dissociate into $H^{\oplus}$ & $OH^{\ominus}$ while a fractional part of D_2O will dissociate into $D^{\oplus}$ & $OD^{\ominus}$. This electrolysis process is completed in seven stages.

$H_2O \rightleftharpoons H^{\oplus} + OH^{\ominus}$

$D_2O \rightleftharpoons D^{\oplus} + OD^{\ominus}$

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$D^{\oplus}$ & $OD^{\ominus}$ due to more mass have less mobility i.e. why $H^{\oplus}$ & $OH^{\ominus}$ will move towards cathode & anode respectively while $D^{\oplus}$ & $OD^{\ominus}$ will be in solution.
This process will be repeated for six times.
Approximately 29000 ltr. of water is electrolysed to get 1 ltr. of 99% pure D_2O.

Properties of Heavy water :

Physical properties :

(a) Heavy water is a colourles, odourless and tasteless mobile liquid.
(b) Nearly all the physical constants are higher than the corresponding values of ordinary water.
(c) Ionic compounds are less soluble in heavy water as compared to ordinary water.

Chemical properties : 6000 parts of normal water contain one part of D_2O. Heavy water is chemically similar to ordinary water. However, D_2O reacts slower than H_2O in chemical reactions.

(a) **Reaction with metals :** D_2O reacts with alkali and alkaline earth metals liberates heavy hydrogen.

$$2D_2O + 2Na \rightarrow \underset{\text{Sodium deuteroxide}}{2NaOD} + D_2$$

$$2D_2O + 2Ca \rightarrow \underset{\text{Calcium deuteroxide}}{Ca(OD)_2} + D_2$$

(b) **Reaction with metallic oxides :** D_2O reacts slowly with basic oxides to form heavy alkalis.

$$Na_2 + D_2O \longrightarrow 2NaOD$$
$$CaO + D_2O \longrightarrow Ca(OD)_2$$

(c) **Reaction with non-metallic oxides :** D_2O reacts slowly with acidic oxides to form deutero acids.

$$N_2O_5 + D_2O \longrightarrow \underset{\text{Deutero nitric acid}}{2DNO_3}$$

$$P_2O_5 + 3D_2O \longrightarrow \underset{\text{Deutero phosphoric acid}}{2D_3PO_4}$$

$$SO_3 + D_2O \longrightarrow \underset{\text{Deutero sulphuric acid}}{D_2SO_4}$$

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(d) **Reaction with metallic carbides, phosphides, nitrides, arsenides, etc. :** Like H_2O heavy water reacts with carbides, phosphide, nitrides, arsenides, etc. to form corresponding deutero compounds.

$$CaC_2 + 2D_2O \longrightarrow Ca(OD)_2 + \underset{\text{Deutero acetylene}}{C_2D_2}$$

$$Al_4C_3 + 12D_2O \longrightarrow 4Al(OD)_3 + \underset{\text{Deutero methane}}{3CD_4}$$

$$Mg_3N_2 + 6D_2O \longrightarrow 3Mg(OD)_2 + \underset{\text{Deutero ammonia}}{2ND_3}$$

$$AlN + 3D_2O \longrightarrow Al(OD)_3 + \underset{\text{Deutero ammonia}}{ND_3}$$

$$Ca_3P_2 + 6D_2O \longrightarrow 3Ca(OD)_2 + \underset{\text{Deutero phosphine}}{2PD_3}$$

$$Na_3As + 3D_2O \longrightarrow 3NaOD + \underset{\text{Deutero arsine}}{AsD_3}$$

(e) **Electrolysis :** A solution of heavy water containing Na_2CO_3 when electrolysed evolve heavy hydrogen at cathode.

$$2D_2O \xrightarrow{Na_2CO_3} 2D_2 + O_2$$

Uses of heavy water(D_2O) :

1. **As a neutron moderator:** Fission in uranium-235 is brought by slow speed neutrons. The substances which are used for slowing down the speed of neutrons are called moderators. Heavy water has high boiling point that's way it is used as moderator as well as coolant in nuclear reactors.
2. For the preparation of Deuterium by electrolysis of D_2O or by reaction of D_2O with Sodium(Na)

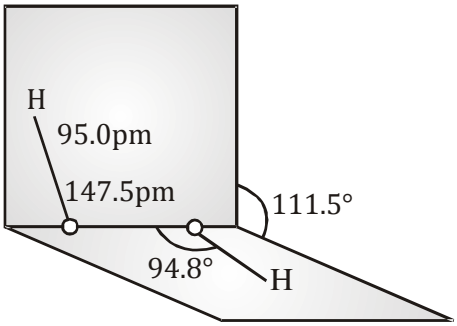
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Hydrogen Peroxide(H_2O_2):

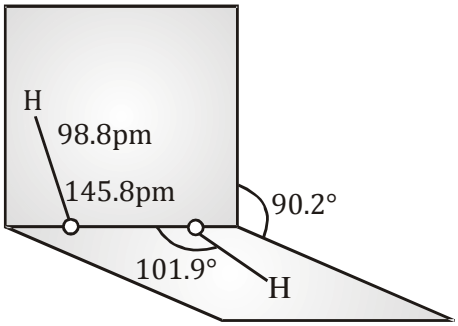
Structure of Hydrogen Peroxide :

The vapour density as determined by Victor Meyer method at 90°C is 17. Hence, the molecular mass of H_2O_2 is 34. Two formulae have been suggested for hydrogen peroxide.

The calculated value of the single bond O–O distance is 1.48Å and X-ray measurements shows that in hydrogen peroxide, O–O bond distance is 147.5 pm. The value of dipole moment of H_2O_2 is 2.1 D. This suggests that all the four atoms do not lie in the same plane. The molecule can be pictured as lying on the spine of a book open to an angle of 111.5°. The hydrogen atoms are present one on each cover and H–O bonds making angles of 94° with the O–O bond as shown in fig. The bond distance between O–H is 0.97Å.



(a) Gas phase

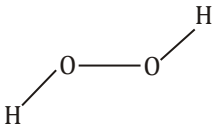


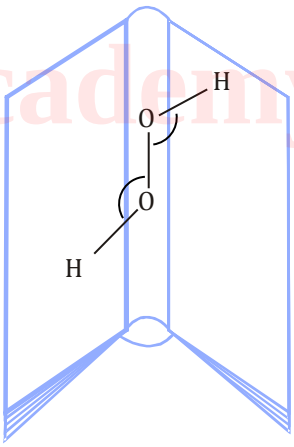
(a) Solid phase

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or





Open book like structure of H_2O_2

Laboratory method :

(i) In laboratory, H_2O_2 is prepared by adding calculated amounts of sodium peroxide to ice cold dilute (20%) solution of H_2SO_4.

$$Na_2O_2 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O_2$$

(ii) By the action of sulphuric acid or phosphoric acid on hydrated barium peroxide $BaO_2.8H_2O$ (Merck process)

(a) $BaO_2.8H_2O + H_2SO_4 \longrightarrow BaSO_4\downarrow + H_2O_2 + 8H_2O$

Anhydrous barium peroxide does not react readily with sulphuric acid (because a coating of insoluble barium sulphate is formed on its surface which stops further action of the acid). Therefore, hydrated barium peroxide, $BaO_2.8H_2O$ must be used.

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(b) $3BaO_2 + 2H_3PO_4 \longrightarrow Ba_3(PO_4)_2 + 3H_2O_2$
$BaO_2 + 2HCl \longrightarrow BaCl_2 + H_2O_2$
$Ba_3(PO_4)_2 + 3H_2SO_4 \longrightarrow 3BaSO_4 + 2H_3PO_4$

Phosphoric acid is preferred over H_2SO_4 because soluble impurities like barium persulphate ($BaO_2.8H_2O - H_2SO_4$) tends to decompose H_2O_2 while H_3PO_4 acts as preservative (negative catalyst) for H_2O_2.

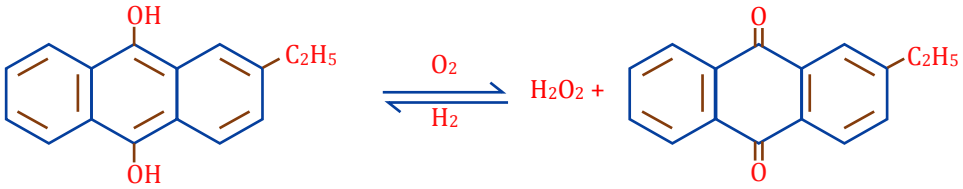
(iii) Peroxodisulphate, obtained by electrolytic oxidation of acidified sulphate solutions at high current density, on hydrolysis yields hydrogen peroxide.

$2HSO_4^-(aq) \xrightarrow{\text{Electrolysis}} HO_3SOOSO_3H(aq) \xrightarrow{\text{Hyrolysis}} 2HSO_4^-(aq) + 2H^+(aq) + H_2O_2(aq)$

This method is now used for the laboratory preparation of D_2O_2.

Industrial Method

(i) **Auto oxidation of 2 ethyl anthraquinol (cyclic process) :**



(ii) **Oxidation of isopropyl alcohol :**



(iii) **Electrolytic Process:** Used 50% H_2SO_4 in electrolytic cell using Pt as anode graphite as cathode.

$2H_2SO_4 \longrightarrow 2H^{\oplus} + 2HSO_4^{\ominus}$ (bisulphate ion)

At Cathode $2H^{\oplus} + 2e^{\ominus} \longrightarrow H_2\uparrow$

At Anode $2HSO_4^{\ominus} \longrightarrow H_2S_2O_8 + 2e^{\ominus}$

$H_2S_2O_8$ = Peroxo disulphhuric acid (Marshall's acid)

$H_2S_2O_8 + H_2O \longrightarrow H_2SO_4 + H_2SO_5$

H_2SO_5 = Peroxo monosulphhuric acid (Caro's acid)

$H_2SO_5 + H_2O \longrightarrow H_2SO_4 + H_2O_2$

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Physical property:

(i) Pure H_2O_2 is colour less, odourless liquid and impure with bluish layer.
(ii) It has more Hydrogen bonding then H_2O, So order of boiling point : $H_2O_2 > D_2O > H_2O$
(iii) It is soluble in H_2O, alcohol and ether
(iv) It has bitter test and harmful for skin
(v) It is a dibasic weak acid
(vi) It has a oxidising as well as reducing property
(vii) H_2O_2 easily decompose in presence of light and temperature, So H_2O_2 always kept in wax lined glass or plastic vessel in dark.
(viii) 30% solution of H_2O_2 is called **Perhydrol**. Which is used as an anti-septic and germicide for washing wounds and ears.

Chemical property:

1. **Stability :** It is unstable in nature and decomposes on standing and heating. It is an example of auto oxidation-reduction Reaction that's why it is always kept in dark or amber coloured or opaque bottles.
$H_2O_2 \longrightarrow H_2O + [O]$ (Nascent oxygen)

Note :

Nascent oxygen working as decolourising agent or bleaching agent(permanent action) .

Decomposition of H_2O_2 ; $2H_2O_2 \rightleftharpoons 2H_2O + O_2$

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Note :

This is retarded by R–OH, acetanilide, glycerol

2. Oxidizing nature : H_2O_2 is a powerful oxidant in acidic as well as in alkaline medium.

(In acid) $H_2O_2 + 2H^{\oplus} + 2e^{\ominus} \xrightarrow{\text{fast}} 2H_2O$

(In alkali) $H_2O_2 + HO^{\ominus} + 2e \xrightarrow{\text{slow}} 3HO^{\ominus}$

Thuse H_2O_2 is more powerful oxidant in acidic medium. The simple interpretation of H_2O_2 as oxidant can be shown by the equation.

$H_2O_2 \longrightarrow H_2O + O$

Following are some important examples of oxidant action of H_2O_2 :

(a) H_2O_2 oxidises black lead sulphide (PbS) to white lead sulphate ($PbSO_4$). This reaction is used in restoring the white colour of old paintings which have blackened due to the formation of lead sulphide by the action of H_2S present in air.

$4[H_2O_2 \longrightarrow H_2O + O]$

$PbS + 4O \longrightarrow PbSO_4$ (white)

$PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$

(b) H_2O_2 oxidises sulphites into sulphates.

$H_2O_2 \longrightarrow H_2O + O$

$Na_2SO_3 + O \longrightarrow Na_2SO_4$

$Na_2SO_3 + H_2O_2 \longrightarrow Na_2SO_4 + H_2O$

(c) H_2O_2 oxidises nitrites to nitrates.

$H_2O_2 \longrightarrow H_2O + O$

$NaNO_2 + O \longrightarrow NaNO_3$

$NaNO_2 + H_2O_2 \longrightarrow NaNO_3 + H_2O$

(d) H_2O_2 oxidises H_2S into sulphur.

$H_2O_2 \longrightarrow H_2O + O$

$H_2S + O \longrightarrow H_2O + S$

$H_2S + H_2O_2 \longrightarrow 2H_2O + S$

3. Reducing nature : It can also act as a reducing agent towards powerful oxidising agents.

$H_2O_2 \longrightarrow 2H^{\oplus} + O_2 + 2e^{\ominus}$

(a) It reduces Ag_2O to silver.

$Ag_2O + H_2O_2 \longrightarrow 2Ag + H_2O + O_2$

(b) It reduces ozone to oxygen.

$H_2O_2 + O_3 \longrightarrow H_2O + 2O_2$

(c) It reduces acidified $K_2Cr_2O_7$ into $Cr_2(SO_4)_3$

$K_2Cr_2O_7 + H_2O_2 + H_2SO_4 \longrightarrow CrO_5 + K_2SO_4 + H_2O \longrightarrow Cr_2(SO_4)_3 + K_2SO_4 + H_2O + O_2\uparrow$

4. Acidic nature : Pure hydrogen peroxide turns blue litmus paper to red indicating its acidic nature. Following are some reaction in which H_2O_2 shows acidic nature:

(a) Reaction with Hydroxides :

$2NaOH + \underset{\text{(Acid)}}{H_2O_2} \longrightarrow \underset{\text{(Sodium peroxide)}}{Na_2O_2} + 2H_2O$

$Ba(OH)_2 + \underset{\text{(Acid)}}{H_2O_2} \longrightarrow \underset{\text{(Barium peroxide)}}{BaO_2} + 2H_2O$

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(b) Reaction with Carbonates :

$$Na_2CO_3 + \underset{\text{(Acid)}}{H_2O_2} \longrightarrow \underset{\text{(Sodium peroxide)}}{Na_2O_2} + 2H_2O + CO_2$$

Precautions :

(i) H_2O_2 cannot be stored in simple glass bottles since rough surface of glass [alkali oxides present in it] excited by light and decomposed H_2O_2.
So the H_2O_2 usually stored in coloured, paraffin wax coated, plastic bottle.

(ii) Always with H_2O_2 add small quantity of inhibitor or negative catalyst to stay decomposition of H_2O_2. like H_3PO_4, acetanilides etc.

Uses

(i) Bleaching agent
(ii) Hair dying
(iii) $H_2O_2 + N_2H_4$ (hydrazine) as Rocket propellent
(iv) H_2O_2 is used as acid, oxidant and reductant
(v) Perhydrol (30% (w/v) or 100V solution of H_2O_2) used as antiseptic and Germicide to clean wounds and ears.

Hard and soft water:

A water is said to be a soft water if it produces sufficient lather with the soap and water is described as being hard if it forms an insoluble scum before it forms a lather with soap. The hardness of natural water is generally caused by presence of bicarbonates, chlorides and sulphates of calcium and magnesium but infact soluble salts that form a scum with soap cause hardness.

$$Ca^{2\oplus}(aq) + 2C_{17}H_{35}COO^{\ominus}(aq) \rightarrow (C_{17}H_{35}COO)_2Ca$$

$$Mg^{2\oplus}(aq) + 2\underset{\text{Anion of soap}}{C_{17}H_{35}COO^{\ominus}}(aq) \rightarrow \underset{\text{Insoluble precipitates}}{(C_{17}H_{35}COO)_2Mg}$$

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Soap will not produce lather with water until all the calcium and magnesium ions have been precipitated. Hard water thus wastes soap. Hardness of water is of two types :

(a) Temporary hardness (b) Permanent hardness

(a) Temporary hardness : This is due to the presence of bicarbonates of calcium and magnesium.

(i) Boiling : Temporary hardness in water is easily removed by boiling, as the bicarbonates decompose readily and the insoluble carbonates are precipitated.

$$Ca(HCO_3)_2 \xrightarrow{\text{Boil}} \underset{\text{(Insoluble)}}{CaCO_3}\downarrow + H_2O + CO_2\uparrow$$

$$Mg(HCO_3)_2 \xrightarrow{\text{Boil}} \underset{\text{(Insoluble)}}{Mg(OH)_2}\downarrow + H_2O + CO_2\uparrow$$

(ii) Clark's process : Temporary hardness can also be removed by Clark's process which involves the addition of slaked lime [$Ca(OH)_2$].

$$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow \underset{\text{(Insoluble)}}{2CaCO_3}\downarrow + 2H_2O$$

$$Mg(HCO_3)_2 + Ca(OH)_2 \rightarrow \underset{\text{(Insoluble)}}{2CaCO_3}\downarrow + \underset{\text{(Insoluble)}}{Mg(OH)_2}\downarrow + 2H_2O$$

It is essential to add only the calculated amount of $Ca(OH)_2$ because excess will cause artificial hardness.

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(b) Permanent hardness : Permanent hardness is introduced when water passes over rocks containing the sulphates or chlorides of both of calcium and magnesium. This type of hardness cannot be removed by boiling or by the addition of slaked lime because the product ($CaCl_2$) of $MgCl_2$ and slaked lime is soluble in water.

$$MgCl_2 + Ca(OH)_2 \rightarrow \underset{\text{(Soluble)}}{CaCl_2} + \underset{\text{(Insoluble)}}{Mg(OH)_2}\downarrow$$

The various water softeners are :

(i) Washing soda : It removes both the temporary and permanent hardness by converting soluble calcium and magnesium compounds into insoluble carbonates.

$$CaCl_2 + NaCO_3 \rightarrow CaCO_3\downarrow + 2NaCl$$

$$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$$

$$Ca(HCO_3)_2 + Na_2CO_3 \rightarrow \underset{\text{(Insoluble)}}{CaCO_3}\downarrow + \underset{\text{(Soluble)}}{2NaHCO_3}$$

$$Mg(HCO_3)_2 + Na_2CO_3 \rightarrow \underset{\text{(Insoluble)}}{MgCO_3}\downarrow + \underset{\text{(Soluble)}}{2NaHCO_3}$$

In place of sodium carbonate, caustic soda or sodium phosphate can also be used.

$$MgCl_2 + 2NaOH \longrightarrow \underset{\text{(Insoluble)}}{Mg(OH)_2}\downarrow + 2NaCl$$

$$3MgSO_4 + 2Na_3PO_4 \longrightarrow \underset{\text{(Insoluble)}}{Mg_3(PO_4)_2}\downarrow + 3Na_2SO_4$$

(ii) Calgon : The polyphosphate complex salt of metaphosphoric acid, in polymeric form $(NaPO_3)_6$, is known as **'Calgon'.** It is represented as $Na_2[Na_4(PO_3)_6]$ (sodium hexametaphosphate). It is also known as Graham's salt. Calcium and Magnesium salts present in hard water react with Calgon to give complex salts.

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$$2CaSO_4 + Na_2[Na_4(PO_3)_6] \longrightarrow Na_2[Ca_2(PO_3)_6] + 2Na_2SO_4$$

$$2MgSO_4 + Na_2[Na_4(PO_3)_6] \longrightarrow Na_2[Mg_2(PO_3)_6] + 2Na_2SO_4$$

(iii) Ion-exchange method by zeolite or Permutit process : Permutit is hydrated sodium alumino silicate.

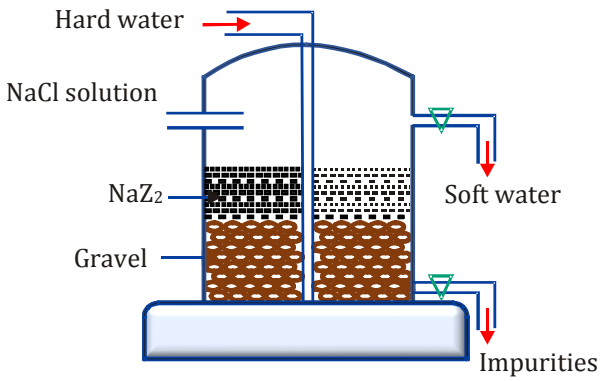
$$[Na_2Al_2Si_2O_8.xH_2O]$$

$$[Na_2O.Al_2O_3.2SiO_2.xH_2O]$$

Permutit also known as sodium zeolite (Na_2Z).

means Zeolite is $Al_2Si_2O_8.xH_2O$.

In this process when hard water is poured into chamber, it may contain organic impurities like plant. This impurities can be removed by gravel.



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On moving upwards hard water will react with Na_2Z during this reaction $Na^{\oplus}$ ions of Na_2Z will replace $Mg^{2\oplus}$ & $Ca^{2\oplus}$ of impurities.

$$Na_2Z + CaCl_2 \longrightarrow 2NaCl + CaZ$$

NaCl is dissolved in water & water becomes soft.

This soft water is fit for washing purpose but not fit for drinking purpose.

After some time when Na_2Z is completely converted into CaZ. Process is stopped and for regeneration of Na_2Z. We use NaCl solution.

$$2NaCl + CaZ \longrightarrow CaCl_2 + Na_2Z$$

(iv) Ion exchange method by synthetic resin : By this process we can remove both cation & anion of hardness.

This process contain two chamber.

(a) Cation exchange resin : This resin contains granular insoluble organic acid having giant molecules with either acidic group ($-SO_3H$ or $-COOH$).

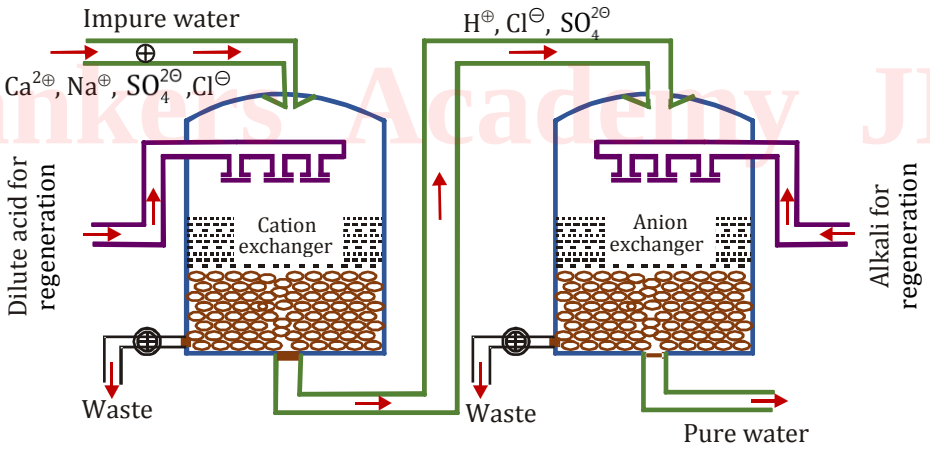
(b) Anion exchange resin : This resin contain giant organic molecules with basic groups derived from amines

$R-NH_3^{\oplus}OH^{\ominus}$ or $OH^{\ominus}$

Process : When hard water is poured into first chamber the cation of hardness ($Mg^{2\oplus}$. $Ca^{2\oplus}$) removed by $H^{\oplus}$ ions of organic acid.

$$2R\text{–}COO^{\ominus}H^{\oplus} + CaCl_2 \longrightarrow (R\text{–}COO)_2Ca + 2H^{\oplus} + 2Cl^{\ominus}$$

$$2R\text{–}COO^{\ominus}H^{\oplus} + MgSO_4 \longrightarrow (R\text{–}COO)_2Mg + 2H^{\oplus} + SO_4^{2\ominus}$$



This water becomes soft but not used for drinking purpose because this water contains the impurity of acid. To remove anion of hardness this acidic water then passed through another bed containing anion exchanger. This exchanger remove anion like $Cl^{\ominus}$, $SO_4^{2\ominus}$ & $HCO_3^{\ominus}$

$$R-NH_3^{\oplus}OH^{\ominus} + H^{\oplus} + Cl^{\ominus} \rightarrow R-NH_3^{\oplus}Cl^{\ominus} + H^{\oplus}OH^{\ominus}$$

$$R-NH_3^{\oplus}OH^{\ominus} + H^{\oplus} + SO_4^{2\ominus} \rightarrow R-NH_3^{\oplus}SO_4^{2\ominus} + H^{\oplus}OH^{\ominus}$$

Regeneration of resin :

(i) Cation exchange resin : We use dil acid.

$$2H^{\oplus}Cl^{\ominus} + (R\text{–}COO)_2Ca \longrightarrow \underset{\text{(Regenerated Resin)}}{2R^{\ominus}COO^{\ominus}H^{\oplus}} + CaCl_2$$

(ii) Anion exchange resin : We use dil NaOH solution

$$R-NH_3^{\oplus}Cl^{\ominus} + Na^{\oplus}OH^{\ominus} \rightarrow Na^{\oplus}Cl^{\ominus} + \underset{\text{(Regenerated Resin)}}{R-NH_3^{\oplus}OH^{\ominus}}$$

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Dihydrogen as A Fuel:

Dihydrogen releases large quantities of heat on combustion. The data on energy released by combustion of fuels like dihydrogen, methane, LPG etc. are compared in terms of the same amounts in mole, mass and volume, are shown in Table.

The Energy Released by Combustion of Various Fuels in Moles, Mass and Volume

Energy released on combustion in kJ liquid state	Dihydrogen (in gaseous)	Dihydrogen (in liquid)	LPG	CH_4 gas	Octane (in state)
per mole	286	285	2220	880	5511
per gram	143	142	50	53	47
per litre	12	9968	25590	35	34005

From this table it is clear that on a mass for mass basis dihydrogen can release more energy than petrol (about three times). Moreover, pollutants in combustion of dihydrogen will be less than that of petrol. The only pollutants will be the oxides of dinitrogen (due to the presence of dinitrogen as impurity with dihydrogen). This, of course, can be minimised by injecting a small amount of water into the cylinder to lower the temperature so that the reaction between dinitrogen and dioxygen may not take place. However, the mass of the containers in which dihydrogen will be kept must be taken into consideration. A cylinder of compressed dihydrogen weighs about 30 times as much as a tank of petrol containing the same amount of energy. Also, **dihydrogen gas is converted into liquid state by cooling to 20K**. This would require expensive insulated tanks. Tanks of metal alloy like $NaNi_5$, Ti–TiH_2, Mg–MgH_2 etc. are used for storage of dihydrogen in small quantities. These limitations have prompted researchers to search for alternative techniques to use dihydrogen in an efficient way.

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In this view **Hydrogen Economy** is an alternative. The basic principle of hydrogen economy is the transportation and storage of energy in the form of liquid or gaseous dihydrogen. Advantage of hydrogen economy is that energy is transmitted in the form of dihydrogen and not as electric power. It is for the first time in the history of India that a pilot project using dihydrogen as fuel was launched in October 2005 for running automobiles. Initially 5% dihydrogen has been mixed in CNG for use in four-wheeler vehicles. The percentage of dihydrogen would be gradually increased to reach the optimum level.

Nowadays, it is also used in fuel cells for generation of electric power. It is expected that economically viable and safe sources of dihydrogen will be identified in the years to come, for its usage as a common source of energy.

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